

What a Derivative Really Is

The single most important reframe in the book, and it happens on page one.

WHERE WE ARE

You have met derivatives before. This chapter is not going to reteach you the power rule. It is going to replace the definition you were given in school — *the derivative is a slope* — with a different one that survives everything this book is going to do to it: curved spacetime, infinite-dimensional function spaces, and eventually differentiation with respect to an entire field configuration.

The replacement is this: **the derivative is the coefficient of the best linear approximation to a function near a point**. Slope is what that means in one dimension on a flat page. It is a special case, and it is the special case that does not generalize. Get the general one now and Chapters 0.6, 2.4, 3.2 and 5.5 become far easier than they have any right to be.

TOOLS YOU'LL NEED — None. This is the ground floor.

1 · The question that forces the idea

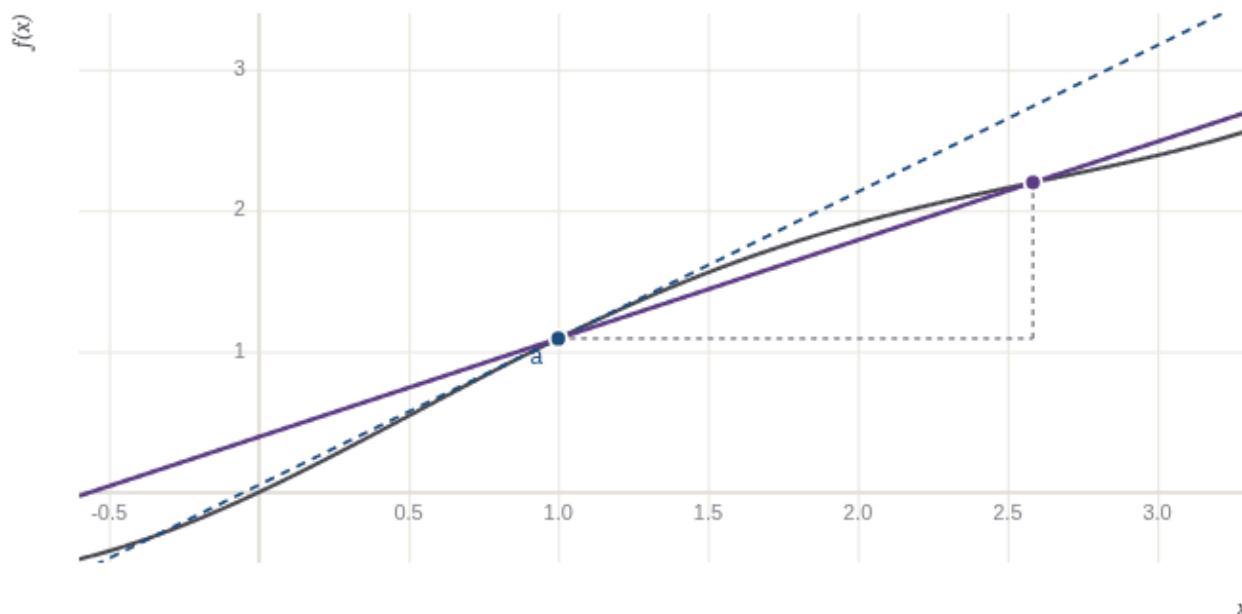
Physics is almost entirely the study of things that change. So the first thing we need is a way to say *how fast*. That sounds trivial until you try to be precise about it.

Suppose a particle's position is $x(t)$. Between times t and $t + h$ it moves a distance $x(t + h) - x(t)$, so its *average* velocity over that stretch is

$$\bar{v} = \frac{x(t + h) - x(t)}{h}. \quad (0.1.1)$$

No controversy there. The trouble starts when you ask for the velocity *right now*, at the single instant t . Set $h = 0$ in (0.1.1) and you get $0/0$, which is not a number. But an object clearly does have a speed at each moment — your speedometer reads something. So there is a real quantity here that the naive algebra cannot reach.

The resolution is to refuse to set $h = 0$ and instead ask what the ratio *approaches* as h shrinks. That evasive-sounding manoeuvre is the whole of differential calculus, and the reason it works is that the numerator and denominator both go to zero *at the same rate*, so their ratio settles down to something finite.



separation h

 $h = 1.585$ $\Delta f / \Delta x = 0.69868 \rightarrow f'(a) = 1.04030$

The secant becomes the tangent. The chord (purple) joins the two marked points on $f(x) = \sin x + \frac{1}{4}x^2$; its slope is the average rate of change over that interval. Drag h toward zero and watch the chord swing onto the blue tangent line while the measured ratio settles on a definite number. Nothing dramatic happens at any particular h — the number simply *converges*. That convergence is the entire content of the definition below.

2 · The definition, stated carefully

The **derivative** of f at the point a is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \quad (0.1.2)$$

whenever that limit exists. The phrase "the limit exists" is doing real work and is not a formality. It means: there is a single number L such that the quotient can be forced as close to L as you like by making h small enough — and crucially, the *same* L whether you approach from the left or the right.

△ WHY THIS ISN'T OBVIOUS

A function can be perfectly continuous — no jumps, no gaps, you can draw it without lifting your pen — and still have no derivative at a point. $f(x) = |x|$ at $x = 0$ is the standard example: approach from the right and the quotient is $+1$, from the left it is -1 , so there is no single limit and no derivative. The graph has a *corner*, and "the best straight line through here" is genuinely ambiguous.

Worse is possible. Weierstrass constructed a function that is continuous *everywhere* and differentiable *nowhere* — infinitely crinkled at every scale, so that zooming in never straightens it out. Such functions are not exotic curiosities; the path of a quantum particle in the Feynman path integral (Chapter 5.8) is exactly of this type. Physics assumes smoothness constantly and quietly, and the places where the assumption fails — shocks, phase transitions, singularities — are reliably the interesting places.

3 · The reframe: derivative as linear approximation

Now rearrange (0.1.2). Saying the quotient tends to $f'(a)$ is the same as saying the difference between the quotient and $f'(a)$ tends to zero:

$$\frac{f(a+h) - f(a)}{h} - f'(a) = \varepsilon(h), \quad \varepsilon(h) \rightarrow 0 \text{ as } h \rightarrow 0. \quad (0.1.3)$$

Multiply through by h and move things around:

$$\boxed{f(a+h) = \underbrace{f(a)}_{\text{value}} + \underbrace{f'(a)h}_{\text{linear in } h} + \underbrace{h\varepsilon(h)}_{\text{smaller than } h}} \quad (0.1.4)$$

Read (0.1.4) slowly, because it is the sentence this entire book is built on. It says: *near a , the function f is a constant plus a linear function of the displacement, and the error you make by stopping there is not merely small — it is small compared to h itself.*

That last clause is the sharp part. Lots of approximations have small errors. This one has an error that vanishes *faster than the thing you are approximating*. And $f'(a)$ is the unique number with that property: no other choice of coefficient makes the leftover shrink faster than linearly. So we can turn the whole thing around and use it as the definition:

THE DEFINITION TO CARRY FORWARD

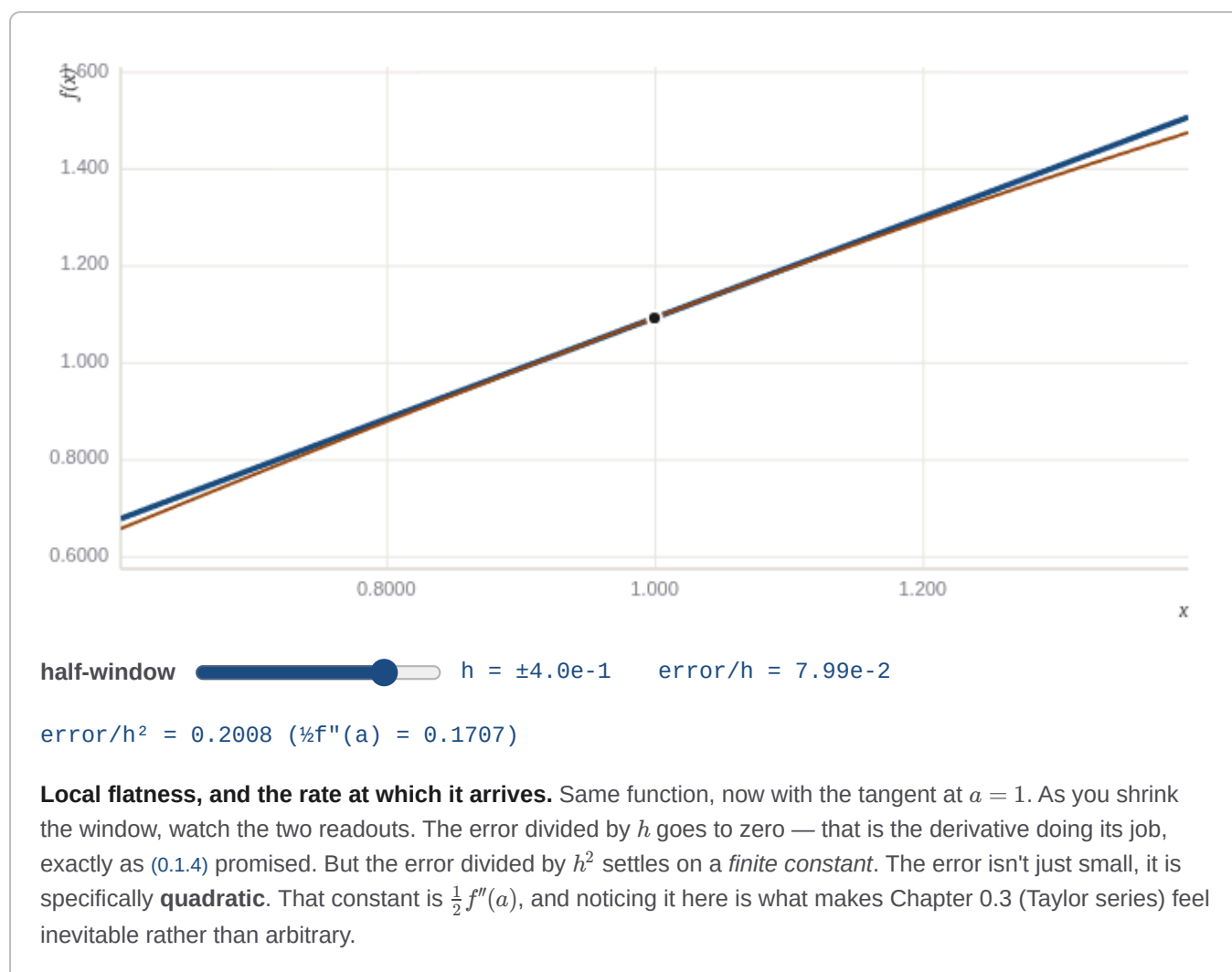
$f'(a)$ is the unique number such that $f(a) + f'(a)h$ approximates $f(a + h)$ with an error that vanishes faster than h .

Nothing in that sentence mentions slopes, graphs, or one-dimensionality. Replace h by a displacement vector and $f'(a)$ by a linear map and you have the derivative in several variables (Chapter 0.6).

Replace the flat displacement by a tangent vector on a curved manifold and you have the derivative in general relativity (Chapter 3.2). Replace the function by a functional and the number by a whole function and you have the variational derivative that produces the Euler–Lagrange equations (Chapter 1.2) and eventually the field equations of the Standard Model. *Every one of those is this same sentence.* "Slope" gets you none of them.

Seeing it: local flatness

There is a way to make (0.1.4) visible. If the error really does die faster than h , then zooming in on the graph near a must make the curve and its tangent line converge — not just look similar, but become indistinguishable in a way that keeps improving the harder you look.



4 · The rules, derived

Almost every differentiation rule you were asked to memorise falls out of (0.1.4) in a line or two. Deriving them once is worth more than memorising them ten times, because the derivations are the ones that generalise.

Sum

Linear approximations add: if $f \approx f(a) + f'(a)h$ and $g \approx g(a) + g'(a)h$, then $f + g \approx (f + g)(a) + (f'(a) + g'(a))h$, and the two error terms are each smaller than h , so their sum is too. Hence $(f + g)' = f' + g'$. Same argument gives $(cf)' = cf'$.

Product — with a picture

Let $A = uv$ and picture it as the area of a rectangle with sides u and v . Nudge x by h : the sides grow by Δu and Δv , and the area grows by three pieces — a strip along the top, a strip along the side, and a tiny corner square:

$$\Delta A = \underbrace{u \Delta v}_{\text{top strip}} + \underbrace{v \Delta u}_{\text{side strip}} + \underbrace{\Delta u \Delta v}_{\text{corner}} \quad (0.1.5)$$

Divide by h . The first two terms give $u \Delta v/h + v \Delta u/h \rightarrow uv' + vu'$. The corner term is $\Delta u \cdot \Delta v/h$, and since $\Delta u \approx u'h$ and $\Delta v \approx v'h$, this is roughly $u'v'h \rightarrow 0$. The corner is *second order* and dies. So

$$(uv)' = u'v + uv'. \quad (0.1.6)$$

The corner square is the entire reason the product rule looks the way it does, and "second-order terms die" is a move you will make hundreds of times before this book is over — most consequentially in Chapter 3.4, where the terms that *refuse* to die when you transport a vector around a closed loop are precisely the curvature of spacetime.

Chain rule — where the reframe pays

Here is the derivation that justifies everything said in §3. We want $\frac{d}{dx} f(g(x))$ at $x = a$. Write $b = g(a)$. Nudge x by h ; then g moves by

$$k \equiv g(a + h) - g(a) = g'(a)h + (\text{smaller than } h). \quad (0.1.7)$$

Now feed that displacement into f , using its own linear approximation at b :

$$\begin{aligned} f(b + k) &= f(b) + f'(b)k + (\text{smaller than } k) \\ &= f(b) + f'(b)g'(a)h + (\text{smaller than } h). \end{aligned} \quad (0.1.8)$$

Compare with (0.1.4) and read off the coefficient of h :

$$(f \circ g)'(a) = f'(g(a)) g'(a). \quad (0.1.9)$$

Notice what actually happened: **we composed two linear approximations, and linear maps compose by multiplication**. That is the whole content of the chain rule. In several variables the linear maps are matrices and they compose by matrix multiplication — the chain rule becomes $D(f \circ g) = Df Dg$, a product of Jacobians (Chapter 0.6). In general relativity it becomes the transformation law that *defines* a tensor (Chapter 2.4). It is the same statement each time. Try getting there from "slope times slope".

▼ Grind box — quotient rule, and the fine print on the chain rule

Quotient. Write $u/v = u \cdot v^{-1}$ and combine the product rule with the chain rule applied to $w \mapsto w^{-1}$, whose derivative is $-w^{-2}$:

$$\left(\frac{u}{v}\right)' = u'v^{-1} + u \cdot (-v^{-2})v' = \frac{u'v - uv'}{v^2}.$$

There is no reason to memorise this separately.

Fine print. The chain-rule argument above has a gap: if $k = 0$ for some sequence of $h \rightarrow 0$ (a perfectly legal thing for g to do — consider g constant), the step " $f(b+k) = f(b) + f'(b)k + o(k)$ " divides by something that may vanish. The clean repair is to define

$$\Phi(k) = \begin{cases} \frac{f(b+k) - f(b)}{k}, & k \neq 0 \\ f'(b), & k = 0 \end{cases}$$

which is continuous at $k = 0$ precisely because f is differentiable there. Then $f(b+k) - f(b) = \Phi(k)k$ holds for *all* k including 0, and

$$\frac{f(g(a+h)) - f(g(a))}{h} = \Phi(k) \cdot \frac{k}{h} \longrightarrow f'(b) g'(a).$$

No division by zero anywhere. This is the kind of hole that does not matter for any function you will meet in physics, but it is worth seeing once that it was noticed and plugged.

5 · Where e comes from

The exponential is not a function someone chose to be fond of. It is forced.

Differentiate b^x from the definition:

$$\frac{d}{dx} b^x = \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} = b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h} \equiv b^x L(b). \quad (0.1.10)$$

Something remarkable is already visible: *whatever* the base, the derivative of b^x is proportional to b^x itself.

Exponentials are the functions whose rate of growth is set by their current size. The constant of proportionality $L(b)$ is an awkward leftover that depends on the base — $L(2) \approx 0.693$, $L(10) \approx 2.303$ — and the natural move is to ask for the base that makes it disappear.

$$e \equiv \text{the unique base with } L(e) = 1, \quad \text{so that} \quad \frac{d}{dx} e^x = e^x. \quad (0.1.11)$$

That is the definition of e : **the base for which the function is its own derivative**. The numerical value $2.71828 \dots$ is a consequence, not a starting point. Setting $L(e) = 1$ means $(e^h - 1)/h \rightarrow 1$, i.e. $e^h \approx 1 + h$ for small h , so $e \approx (1 + h)^{1/h}$ — which for $h = 1/n$ is the familiar $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n$.

And the logarithm follows immediately by the chain rule. From $e^{\ln x} = x$, differentiate both sides: $e^{\ln x} \cdot (\ln x)' = 1$, so $(\ln x)' = 1/x$.

FAMILIAR GROUND

You already use (0.1.11) daily, under a different name. A drug cleared by first-order kinetics obeys

$$\frac{dC}{dt} = -kC \quad \implies \quad C(t) = C_0 e^{-kt}, \quad t_{1/2} = \frac{\ln 2}{k}.$$

The differential equation says exactly what (0.1.10) said: the rate of change is proportional to the amount present. That is why the solution *had* to be an exponential — nothing else has that property. The same three lines describe radioactive decay, a discharging capacitor, and the washout of a tracer.

Now here is the part worth filing away. In Chapter 4.4 the fundamental law of quantum mechanics will turn out to be

$$i\hbar \frac{d}{dt} |\psi\rangle = \hat{H} |\psi\rangle \quad \implies \quad |\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle.$$

Structurally that is your clearance equation. The *only* new ingredient is the factor of i — and in Chapter 0.4 you will see that multiplying by i is a rotation by 90° . So the difference between a drug concentration decaying away and a quantum state evolving in time is that one exponential **shrinks** and the other **rotates**. Same equation, turned through a right angle. Hold onto that; it is not a metaphor.

6 • Trigonometric derivatives

These need one genuinely geometric input. For $0 < h < \pi/2$, comparing the areas of a triangle, a circular sector, and a larger triangle inside the unit circle gives $\sin h < h < \tan h$. Divide through by $\sin h$:

$$1 < \frac{h}{\sin h} < \frac{1}{\cos h} \quad \implies \quad \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad (0.1.12)$$

since $\cos h \rightarrow 1$ and the middle quantity is trapped. This is where the fact that angles are measured in *radians* enters physics permanently — in degrees the limit would be $\pi/180$ and every formula downstream would carry the litter.

▼ Grind box — from the squeeze to $\sin' = \cos$

First a companion limit. Using $1 - \cos h = \frac{\sin^2 h}{1 + \cos h}$,

$$\frac{1 - \cos h}{h} = \frac{\sin h}{h} \cdot \frac{\sin h}{1 + \cos h} \longrightarrow 1 \cdot \frac{0}{2} = 0.$$

Now expand with the angle-addition identity:

$$\begin{aligned} \frac{\sin(x+h) - \sin x}{h} &= \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h} \\ &\longrightarrow \sin x \cdot 0 + \cos x \cdot 1 = \cos x. \end{aligned}$$

The same manipulation on $\cos$ gives $\cos' = -\sin$. Note the consequence: $\frac{d^2}{dx^2} \sin x = -\sin x$. A function proportional to minus its own second derivative — that is the harmonic oscillator equation, and it is about to become the most reused equation in this book (Chapters 0.8, 4.5, 5.3, 7.4).

7 • Higher derivatives, and what they are for

Differentiate again and you get f'' , the rate at which the rate is changing. Physically the first three are so common they have names: position, velocity $\dot{x}$, acceleration $\ddot{x}$. Newton's second law is a statement about the *second* derivative, which is exactly why mechanics is hard: you are told the acceleration and asked for the position, and you must integrate twice, picking up two constants that encode where the thing started and how fast it was going.

Geometrically f'' is concavity, but the sharper statement is the one the zoom figure already showed you numerically: f'' is the *leading error in linearization*,

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + (\text{smaller than } h^2). \tag{0.1.13}$$

Keep going and you get Taylor's theorem, which is Chapter 0.3 and which more or less runs the rest of physics.

Notation, and one deliberate abuse

FORM	WRITTEN	BEST FOR
Lagrange	$f'(x), f''(x)$	Pure functions of one variable
Leibniz	$\frac{df}{dx}, \frac{d^2f}{dx^2}$	Anything where you must track <i>with respect to what</i>
Newton	$\dot{x}, \ddot{x}$	Derivatives with respect to <i>time</i> , universally, in mechanics
Operator	$Df, \partial_x f$	When the derivative itself is the object of study — QM, QFT

⚠ WHY THIS ISN'T OBVIOUS

$\frac{dy}{dx}$ is *not* a fraction. It is a single symbol denoting a limit of fractions, and the limit has already been taken. Yet every physicist you will ever read writes things like $dy = f'(x) dx$ and cheerfully cancels differentials, and they are not being sloppy — they are being *terse about something true*.

The justification is (0.1.4). The derivative is a linear map; dx is an input displacement and dy the corresponding output displacement, and $dy = f'(x)dx$ is a legitimate equation between linear maps, not an approximation. Separating variables in a differential equation works for the same reason. In Chapter 0.6 the object dx becomes a **one-form** and acquires a proper definition; in Chapter 3.5 it acquires an algebra, and $dx \wedge dy = -dy \wedge dx$ starts to matter. So: use the abuse, it is sound — but know that you are cashing a cheque written in Chapter 0.6.

8 · Worked examples

WORKED EXAMPLE 1 — THE LOGARITHMIC DERIVATIVE

A pendulum has period $T = 2\pi\sqrt{L/g}$. You lengthen it by 1%. What happens to the period?

You could compute T twice and subtract. Don't. Take the logarithm first, which converts the product-and-power structure into a sum:

$$\ln T = \ln 2\pi + \frac{1}{2} \ln L - \frac{1}{2} \ln g.$$

Now differentiate. Since $(\ln u)' = u'/u$, each term becomes a *fractional* change:

$$\frac{dT}{T} = \frac{1}{2} \frac{dL}{L} - \frac{1}{2} \frac{dg}{g}.$$

With g fixed and $dL/L = 1\%$, we get $dT/T = 0.5\%$. The pendulum runs half a percent slow.

The general rule, worth having permanently: for $f = u^a v^b / w^c$,

$$\frac{df}{f} = a \frac{du}{u} + b \frac{dv}{v} - c \frac{dw}{w}.$$

Exponents become sensitivity coefficients. This one identity does most error-propagation work in experimental physics, and it is why a quantity's exponent tells you at a glance how much it matters. It is the linearization idea of §3 applied to $\ln f$ instead of f .

WORKED EXAMPLE 2 — WHERE $\frac{1}{2}MV^2$ ACTUALLY COMES FROM

In Chapter 2.5 the energy of a moving particle will turn out to be $E = \gamma mc^2$ with

$$\gamma(v) = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}.$$

Nothing about that looks like the kinetic energy you know. Linearize it. Set $u = -v^2/c^2$ and apply the chain rule to $\gamma = (1 + u)^{-1/2}$:

$$\frac{d\gamma}{dv} = -\frac{1}{2}(1 + u)^{-3/2} \cdot \frac{du}{dv} = -\frac{1}{2}(1 + u)^{-3/2} \cdot \left(-\frac{2v}{c^2}\right) = \frac{v}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2}.$$

At $v = 0$ this is zero, so the linear term vanishes and we need the quadratic one. Differentiate again and evaluate at $v = 0$: only the first factor survives, giving $\gamma''(0) = 1/c^2$. Feeding that into (0.1.13) with $a = 0$:

$$\gamma(v) \approx 1 + \frac{1}{2} \frac{v^2}{c^2} \quad \Longrightarrow \quad E = \gamma mc^2 \approx \underbrace{mc^2}_{\text{rest energy}} + \underbrace{\frac{1}{2}mv^2}_{\text{kinetic energy}}.$$

The kinetic energy of your first physics course is *the quadratic term in the linearization of relativity*. It was never a fundamental quantity; it was the second-order coefficient, and the $\frac{1}{2}$ that seemed arbitrary is the $\frac{1}{2}$ sitting in (0.1.13). This is what it feels like when a new theory contains an old one: you expand, and the old theory is the first surviving term.

9 · Your turn

PROBLEM 1 — INVERSE FUNCTIONS

Without using the formula for $(\ln x)'$, show that if f is differentiable and invertible then $(f^{-1})'(y) = 1/f'(f^{-1}(y))$. Then use it to get $(\ln x)'$ and $(\arcsin x)'$.

– Solution

Differentiate the identity $f(f^{-1}(y)) = y$ with respect to y using the chain rule:

$$f'(f^{-1}(y)) \cdot (f^{-1})'(y) = 1 \implies (f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Geometrically: reflecting a graph in the line $y = x$ turns a slope into its reciprocal.

Logarithm. $f = e^x$, $f' = e^x$, $f^{-1} = \ln y$, so $(\ln y)' = 1/e^{\ln y} = 1/y$.

Arcsine. $f = \sin x$, $f' = \cos x$, so $(\arcsin y)' = 1/\cos(\arcsin y)$. Since $\cos \theta = \sqrt{1 - \sin^2 \theta}$ on the principal branch and $\sin(\arcsin y) = y$,

$$(\arcsin y)' = \frac{1}{\sqrt{1 - y^2}}.$$

That $\sqrt{1 - y^2}$ is not a coincidence — it is the same square root that appears in γ , and for the same geometric reason.

PROBLEM 2 — THE ERROR REALLY IS QUADRATIC

Linearize e^x at $x = 0$, i.e. approximate it by $1 + x$. Compute the exact error $E(h) = e^h - (1 + h)$ for $h = 10^{-1}, 10^{-2}, 10^{-3}$ and verify numerically that $E(h)/h \rightarrow 0$ while $E(h)/h^2 \rightarrow \frac{1}{2}$. Explain the $\frac{1}{2}$.

– Solution

h	$E(h)$	E/h	E/h^2
10^{-1}	5.171×10^{-3}	5.17×10^{-2}	0.5171
10^{-2}	5.017×10^{-5}	5.02×10^{-3}	0.5017
10^{-3}	5.002×10^{-7}	5.00×10^{-4}	0.5002

E/h marches to zero, confirming that the linear approximation is correct in the strong sense of (0.1.4). E/h^2 marches to $\frac{1}{2}$ instead, and by (0.1.13) that limit must be $\frac{1}{2}f''(0) = \frac{1}{2}e^0 = \frac{1}{2}$. ✓

The moral: "the error is small" is a weak statement. "The error is $O(h^2)$ with known coefficient" is a usable one, and it is what lets you decide in advance how many terms you need.

PROBLEM 3 — SENSITIVITY IN PRACTICE

The escape velocity from a body of mass M and radius R is $v = \sqrt{2GM/R}$. Using the logarithmic-derivative rule, estimate the percentage change in v if M is 3% larger and R is 2% larger. Which quantity is v more sensitive to?

– Solution

$v = \sqrt{2G} M^{1/2} R^{-1/2}$, so

$$\frac{dv}{v} = \frac{1}{2} \frac{dM}{M} - \frac{1}{2} \frac{dR}{R} = \frac{1}{2}(3\%) - \frac{1}{2}(2\%) = +0.5\%.$$

Equally sensitive in magnitude to both — the exponents are $\pm\frac{1}{2}$ — but opposite in sign, so the two effects largely cancel. Note how much faster this is than evaluating v twice, and that it also tells you something the two evaluations wouldn't: *why* they nearly cancel.

The linearization is valid because 3% is small; at 300% it would be worthless, and you would need the higher terms of Chapter 0.3.

PROBLEM 4 — HALF-LIFE VERSUS TIME CONSTANT

For $C(t) = C_0 e^{-kt}$, find the time to fall to $1/e$ of the initial value and the time to fall to $1/2$. Show the ratio is $1/\ln 2$ and compute it. Then: a colleague says "the drug is 63% cleared after one half-life." Where did they go wrong, and what is the correct statement?

– Solution

$e^{-k\tau} = e^{-1}$ gives the time constant $\tau = 1/k$. $e^{-kt_{1/2}} = \frac{1}{2}$ gives $t_{1/2} = \ln 2/k \approx 0.693/k$.

Ratio: $\tau/t_{1/2} = 1/\ln 2 \approx 1.443$, so the time constant is about 44% longer than the half-life.

The colleague has fused the two. After one *time constant* τ , the fraction remaining is $1/e \approx 0.368$, so 63.2% is gone. After one *half-life*, by definition, 50% is gone. The 63% figure belongs to τ , not $t_{1/2}$, and the two differ by that factor of 1.443.

Why the confusion is so persistent: τ is the natural constant of the differential equation — it is what appears in $\frac{dC}{dt} = -C/\tau$ — while $t_{1/2}$ is the natural constant for a human reading a chart. The mathematics prefers e ; our base-ten intuitions prefer halves; and $\ln 2$ is the toll for crossing between them.

THE BRICK YOU JUST LAID

You now have the derivative in the form that generalizes: $f(a+h) = f(a) + f'(a)h + o(h)$ — the coefficient of the best linear approximation, with an error that dies faster than linearly. You have the rules as consequences rather than incantations, with the chain rule understood as *composition of linear maps*. You have e defined by the property that makes it inevitable, and you have seen that quantum time evolution is your own first-order kinetics with an i in the exponent.

Where this gets spent. The linear-approximation view is cashed in Chapter 0.6 (the total derivative as a matrix), Chapter 1.2 (varying a whole path instead of a number), Chapter 3.2 (tangent spaces on a curved manifold), and Chapter 5.5 (differentiating with respect to a field). The observation that the leftover error is *quadratic* is the door into Chapter 0.3. And $\sin'' = -\sin$ — a function proportional to minus its own second derivative — is the harmonic oscillator, which from Chapter 0.8 onward never leaves.